

Unfolding of relative g -entropies and monotone metrics

Fabio Di Nocera

Max Planck-Institut für Mathematik in den Naturwissenschaften, Leipzig, Germany

July 22, 2022

MaxEnt'22

- 1 Introduction, some notions of Quantum IG
- 2 An unfolding procedure for the space of quantum states
- 3 Quantum IG on the unfolding space

Space of (mixed) Quantum States

$\mathcal{H}$ finite-dimensional Hilbert space (complex dimension n), so $\mathcal{B}(\mathcal{H})$ is basically made of $(n \times n \text{ complex})$ matrices.

A **quantum state** ρ is a self-adjoint matrix whose eigenvalues are non-negative and sum to 1,

$$\overline{\mathcal{S}(\mathcal{H})} = \{\rho \in \mathcal{B}(\mathcal{H}) : \rho \geq 0 \quad \text{and} \quad \text{Tr}(\rho) = 1\}. \quad (1)$$

$\overline{\mathcal{S}(\mathcal{H})}$ is called the **space of quantum states**.

A **faithful quantum state** ρ is a self-adjoint matrix whose eigenvalues are (strictly!) positive and sum to 1,

$$\mathcal{S}(\mathcal{H}) = \{\rho \in \mathcal{B}(\mathcal{H}) : \rho > 0 \quad \text{and} \quad \text{Tr}(\rho) = 1\}. \quad (2)$$

$\mathcal{S}(\mathcal{H})$ is called the **space of faithful quantum states**.

Information metrics

In Quantum Information Geometry we use completely-positive trace-preserving (**CPTP**) maps¹.

There is not a unique choice for monotone metrics in the quantum case.

A general characterization for information metric has been given by Petz.

This characterization establishes a one-to-one correspondence between operator monotone functions² f such that

$$t f(t^{-1}) = f(t) \quad \text{and} \quad f(1) = 1 \quad (3)$$

and monotone metrics G^f .

¹A linear map $\phi : \mathcal{M}_n(\mathbb{C}) \rightarrow \mathcal{M}_m(\mathbb{C})$ is CP if $\phi \otimes id_k$ from $\mathcal{M}_n(\mathbb{C}) \otimes \mathcal{M}_k(\mathbb{C})$ to $\mathcal{M}_m(\mathbb{C}) \otimes \mathcal{M}_k(\mathbb{C})$ is positive for every k .

² f is monotone w.r.t. the partial order on $\mathcal{M}_n(\mathbb{C})$ given by: $A > B$ if $A - B$ is positive definite

Relative g -entropies and information metrics

Relative g -entropies are defined as two-point functions on the space of quantum states,

$$H_g : \mathcal{S}(\mathcal{H}) \times \mathcal{S}(\mathcal{H}) \ni (\rho, \sigma) \mapsto S_g(\rho, \sigma) = \text{Tr} \left(\sqrt{\rho} g(\mathbf{L}_\sigma \mathbf{R}_{\rho^{-1}}) (\sqrt{\rho}) \right) \in \mathbb{R}, \quad (4)$$

where g is an operator convex function.

Monotonicity condition:

$$H_g^{\mathcal{H}}(\rho, \sigma) = H_g^{\mathcal{K}}(\Phi(\rho), \Phi(\sigma)) \quad (5)$$

For all CPTP maps $\Phi : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{K})$

A sort of second-order expansion of these entropies gives a monotone metric.

The big picture

$$f \longrightarrow G_f$$

$$g \longrightarrow H_g \longrightarrow G_g$$

If f and g are related as

$$f(x) = \frac{(1-x)^2}{g(x) + xg(x^{-1})}. \quad (6)$$

Then

$$\begin{array}{ccc} f & \longrightarrow & G \\ & & \nearrow \\ g & \longrightarrow & H_g \end{array}$$

Quantum States and Probability distributions

There exists a basis $\{|e_1\rangle, |e_2\rangle, \dots, |e_n\rangle\}$ s.t.

$$\rho = \sum_j \lambda_j |e_j\rangle \langle e_j|. \quad (7)$$

Change of basis are made by means of unitary operators! This implies that $\exists U \in \mathcal{U}(\mathcal{H}) : \rho = U\rho_0 U^\dagger$ with

$$\rho_0 = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix} \quad (8)$$

Notice that $\rho > 0$ and $\text{Tr}(\rho) = 1$ imply that $(\lambda_1, \lambda_2, \dots, \lambda_n)$ is a **probability distribution**.

Quantum States and Probability distributions

$$i : \Delta_{n-1}^{\circ} \ni \mathbf{p} = (p_1, \dots, p_n) \longrightarrow i(\mathbf{p}) = \text{diag}(p_1, \dots, p_n) \in \mathcal{S}(\mathcal{H}). \quad (9)$$

Then for any quantum state ρ we have

$$\exists U \in \mathcal{U}(\mathcal{H}) \quad \text{and} \quad \mathbf{p} \in \Delta_{n-1}^{\circ} : \rho = Ui(\mathbf{p})U^{\dagger}. \quad (10)$$

Unfolding:

$$\pi : \mathcal{M}(\mathcal{H}) := \mathcal{U}(\mathcal{H}) \times \Delta_{n-1}^{\circ} \ni (U, \mathbf{p}) \mapsto Ui(\mathbf{p})U^{\dagger} \in \mathcal{S}(\mathcal{H}). \quad (11)$$

Dequantization map:

$$\pi_D : \mathcal{M}(\mathcal{H}) := \mathcal{U}(\mathcal{H}) \times \Delta_{n-1}^{\circ} \ni (U, \mathbf{p}) \mapsto \mathbf{p} \in \Delta_{n-1}^{\circ}. \quad (12)$$

Tangent vectors

$\mathcal{M}(\mathcal{H})$ is the product of two manifolds,

$$\mathcal{M}(\mathcal{H}) = \mathcal{U}(\mathcal{H}) \times \Delta_{n-1}^{\circ}. \quad (13)$$

Its tangent space at $(U, \mathbf{p})$ is given by

$$T_{(U, \mathbf{p})}\mathcal{M}(\mathcal{H}) = T_U\mathcal{U}(\mathcal{H}) \times T_{\mathbf{p}}\Delta_{n-1}^{\circ}, \quad (14)$$

implying that $V_{(U, \mathbf{p})} \in T_{(U, \mathbf{p})}\mathcal{M}(\mathcal{H})$ can be written as

$$V_{(U, \mathbf{p})} = (iH, v), \quad (15)$$

with H a self-adjoint matrix and $v \in \mathbb{R}^n$ such that $\sum_j v^j = 0$.

Information metrics on the unfolding space

We will now pull-back this metric tensor to the unfolding space,

$$G_{\mathcal{M}}^f := \pi^* G^f, \quad (16)$$

meaning that

$$\begin{aligned} \left(G_{\mathcal{M}}^f \right)_{(U, \mathbf{p})} ((iH, v), (iK, u)) &:= \left(\pi^* G^f \right)_{(U, \mathbf{p})} ((iH, v), (iK, u)) \\ &= (G^f)_{\pi(U, \mathbf{p})} (T_{(U, \mathbf{p})} \pi(iH, v), T_{(U, \mathbf{p})} \pi(iK, u)). \end{aligned} \quad (17)$$

Here $T_{(U, \mathbf{p})} \pi$ is the tangent map of π at the point $(U, \mathbf{p})$, H and K are self-adjoint operators on $\mathcal{H}$ and $u, v \in T_{\mathbf{p}} \Delta_{n-1}^{\circ}$.

Information metrics on the unfolding space

It can be shown that

$$\left(G_{\mathcal{M}}^f\right)_{(U, \mathbf{p})}((iH, v), (iK, u)) = G_{\mathcal{U}}^f(iH, iK) + \pi_D^* \left(G_{FR}^{n-1}(v, u)\right). \quad (18)$$

Some observations on this expression:

- The last term of this expression can be considered as some sort of classical term of the metric $G_{\mathcal{M}}^f$.
- This term does not depend on the choice of the operator monotone function f .
- The term $G_{\mathcal{U}}^f(iH, iK)$, depends on the choice of the operator monotone function f .
- It can be shown that this term is equal to zero whenever H and K commute with $\pi(U, \mathbf{p})$, this for all f .

Classical Information Geometry can in a sense always be considered as contained in Quantum Information Geometry.

Relative g -entropies on the unfolding space

In order to pull-back two-point functions, we need to define a “double projection”, i.e.

$$\Pi : ((U, \mathbf{p}), (V, \mathbf{q})) \mapsto (\pi(U, \mathbf{p}), \pi(V, \mathbf{q})). \quad (19)$$

Now we can pull-back relative g -entropies,

$$\Pi^* (S_g^{\mathcal{H}}) ((U, \mathbf{p}), (V, \mathbf{q})) = \sum_{j,k} g \left(\frac{q^j}{p_k} \right) \langle e_k | U^\dagger V | e_j \rangle \langle e_j | V^\dagger U | e_k \rangle. \quad (20)$$

Clearly the second derivatives will now be carried out using the tools of differential geometry and in particular, for the part containing the unitary operators, making use of the differential calculus on Lie groups.

From relative g -entropies to monotone quantum metrics: an unfolded version

Taking second derivatives of the entropy one gets

$$(G_{\mathcal{M}}^g)_{(U, \mathbf{p})}((iH, v), (iK, u)) = G_{\mathcal{U}}^g(iH, iK) + \pi_D^* (G_{FR}^{n-1}(v, u)).$$

Comparing this expression with the one given by Petz characterization,

$$f(x) = \frac{(1-x)^2}{g(x) + xg(x^{-1})}. \quad (21)$$

Which in turn also implies $g(1) = 0$ and $g''(1) = f(1) = 1$.

This is in complete agreement with the situation on the space of quantum states.

Conclusions and outlooks

- We described the relation between operator monotone functions, operator convex functions, monotone metrics and relative g -entropies.
- The same relation holds in the unfolding space!
- Computations in the unfolding space are simplified by standard results about differential calculus on Lie groups.
- The unfolding procedure allows to identify the classical and non-classical contributions on quantities that are relevant from an information-theoretic point of view.

Thank You for Your attention!

Backup slide 1: Stratification of the Space of Quantum States

$\overline{\mathcal{S}}_n$ is **not** a manifold, the following results hold:

Theorem

The spaces $\mathcal{S}_n^k$ of states of rank k , with $1 \leq k \leq n$ are smooth and connected manifolds in $\mathcal{V}$ of real dimension $2nk - k^2 - 1$.

Corollary

The boundary $\partial\mathcal{S}_n$ is the union of all strata with $k < n$. It is not a smooth manifold unless $n = 2$.

Moreover $\mathcal{S}_n \equiv \mathcal{S}_n^n$ is an open set.

Backup slide 2: Computations relative to relative g -entropies and monotone metrics on the unfolding space

$$\begin{aligned}
 G_g^{\mathcal{M}}(X, Y) &= g''(1) \pi_r^* G_{FR}^{n-1}(X, Y) \\
 &+ g(1) \sum_{j=1}^n p^j \langle e_j | U^\dagger dU(X) U^\dagger dU(Y) + U^\dagger dU(Y) U^\dagger dU(X) | e_j \rangle \\
 &- 2g(1) \sum_{j=1}^n p^j \langle e_j | U^\dagger dU(X) | e_j \rangle \langle e_j | U^\dagger dU(Y) | e_j \rangle \\
 &- 2 \sum_{k>j} \left(g \left(\frac{p^j}{p^k} \right) p^k + g \left(\frac{p^k}{p^j} \right) p^j \right) \\
 &\cdot \operatorname{Re} \left(\langle e_k | U^\dagger dU(X) | e_j \rangle \langle e_j | U^\dagger dU(Y) | e_k \rangle \right).
 \end{aligned} \tag{22}$$

$$G_g^{\mathcal{M}} = \pi_D^* G_{FR}^{n-1}$$

$$-2 \sum_{k>j} (p_k - p_j)^2 \left(p_k f \left(\frac{p^j}{p^k} \right) \right)^{-1} \operatorname{Re} (\langle e_k | \omega | e_j \rangle \otimes \langle e_j | \omega | e_k \rangle).$$

Backup slide: Petz characterization

Petz characterized all metric tensors on the space of quantum states obeying monotonicity under CPTP maps,

$$G^f|_{\rho}(A, B) = \text{Tr} \left(A \left(\mathbf{K}_{\rho}^f \right)^{-1} (B) \right). \quad (23)$$

$\mathbf{K}_f$ is a superoperator defined as

$$\mathbf{K}_{\rho}^f = f(\mathbf{L}_{\rho} \mathbf{R}_{\rho^{-1}}) \mathbf{R}_{\rho}, \quad (24)$$

$\mathbf{L}_{\rho}$ and $\mathbf{R}_{\rho}$ are respectively the left and right multiplications by means of ρ ,

$$\mathbf{L}_{\rho}(B) = \rho B, \quad \mathbf{R}_{\rho}(B) = B \rho. \quad (25)$$